

CHAPTER 12 From the Book by Robert C. Wilson, AL7KK Economical Antennas for Low & Medium Frequencies

12.0 The Vertical Antenna and Some Variations

12.1 The Basic Vertical Antenna. The most basic low frequency antenna is the quarter wavelength vertical antenna. A quarter wave is practical and reasonable above about 1000 kHz. Below roughly 1000 kHz, the vertical requires considerably greater economic resources. Since this book is about economic antennas, I shall try to concentrate on antennas less than approximately 150 feet, or 50 meters, in height.

The most basic tubular or lattice work quarter wave vertical is simply one half of a half wave dipole antenna. It uses the ground as a mirror to simulate the missing half of the dipole. However using the ground as a mirror requires a low resistance ground system. Low losses mean using salt-water grounds, or extensive wire grounds. Much of this has already been covered in Chapter 5.

Vertical antennas tend to radiate along the surface of the earth giving the best distance coverage for the money. There will be a notch in the lowest part of the vertical pattern of all vertical antennas. This will occur unless the ground is perfect, like a sheet of copper over the ground. The depth of this notch will greatly effect range.

Because the vertical is the basic antenna from which almost all other low frequency antennas are derived I will spend a bit of time showing some details of such antennas operating between 100 and 2000 Khz. Table 12.1-1 shows the impedances for four heights of tower, tuner parameters, and estimated range to the 70-microvolt contour using a 1000 Watt transmitter.

Notice that shorter antennas at low frequencies have lower radiation resistances and consequently lower efficiency. Low efficiency means

shorter ranges. A 300 meter (1000 feet) antenna is very expensive, but a 42 meter (140 foot) antenna is much cheaper, easier to transport and faster to construct. Low cost is important, so I have looked for ways to attempt to make short antennas more efficient. Here I have attempted to make them reasonably competitive with taller straight verticals by explaining how to raise their efficiency. Still it is up to your budget ability to select the necessary antenna, and there are quite a range of selections implied in Table 12.1-1 and in the antenna descriptions that follow.

Since the emphasis of this book is on low cost antennas, the antennas tend to be the smallest reasonable diameters. The mast-type vertical antennas have a diameters from 3 meters (10 feet) for the largest, 2 meters (6.5 feet) for the next smaller, 1 meter (3.28 feet) for the next smaller, and 0.3 meter (1 foot) for the smallest tower.

Long field experience indicates that about 5 Ohms may be added to the calculated value of antenna radiation resistance in the table due to combined losses in the antenna and ground system. That assumes a fairly good ground system. The higher the radiation resistance, the better the antenna efficiency and the longer the station range.

It is well known that antennas with a height of $5/8$ wavelength will probably be optimum, but at low frequency, antenna cost is often the primary limiting station factor and such an antenna is very expensive.

In Table 12.1-1 frequencies are given in Kilohertz, heights and diameters in meters, radiation resistance and reactance in Ohms. Capacitive reactance is indicated by a negative sign (-), and inductive reactance by a plus sign (+).

Freq. kHz	Ht. m.	Dia. m.	R Ohm	X Ohm	Tune form	Tune C pF.	Tune L uH.	eff. % est.	Range nt. mi. 1 KW.
100	300	3	3.5	-2771		32,800	518	41%	740
200	300	3	21	-55	1	25,670	68	81%	814
400	300	3	475	-9	1	5,309	169	99%	757
100	150	2	0.75	-562	1	31,055	976	13%	465
200	150	2	3.3	-250	1	16,340	238	40%	652
400	150	2	21.	-49	1	12,835	32	81%	716
1000	150	2	86.	-199	1	5,481	5.4	95%	628
100	75	1	0.18	-1152	1	30,713	1,916	3.5%	227
200	75	1	0.75	-562	1	15,377	488	13%	421
400	75	1	3.3	-250	1	8,170	119	40%	578
1000	75	1	48	+23	2	4,974	1.43	91%	628
100	42	0.3	0.06	-2546	1	30,642	4,135	1.2%	105
200	42	0.3	0.24	-1263	1	15,377	1,046	4.6%	249
300	42	0.3	0.58	-830	1	10,317	468	10%	353
400	42	0.3	1.0	-611	1	7,802	263	17%	425
500	42	0.3	1.7	-476	1	6,328	168	25%	474
600	42	0.3	2.4	-384	1	5,348	115	32%	501
700	42	0.3	3.4	-315	1	4,678	83	41%	528
800	42	0.3	4.6	-262	1	4,197	62	48%	541
900	42	0.3	6.1	-218	1	3,853	47	55%	551
1000	42	0.3	7.9	-181	1	3,609	38	61%	557
1100	42	0.3	10.	-149	1	3,445	28	67%	562
1200	42	0.3	13.	-120	1	3,401	21	72%	564
1300	42	0.3	16.	-94	1	3,401	15.9	76%	564
1400	42	0.3	20.	-69	1	3,553	11.5	80%	563
1500	42	0.3	24.	-44	1	3,789	7.64	83%	561
1600	42	0.3	30.	-21	1	4,521	4.28	86%	559
1700	42	0.3	37	+2.7	2	5,289	1.48	88%	556
1800	42	0.3	46	+2.7	2	2,679	0.531	90%	553
1900	42	0.3	58.	+51	3	15,603	0.503	92%	551
2000	42	0.3	72	+76	3	5,062	1.59	94%	548

Table 12.1-1. Basic Vertical Antennas from 42 meters (140 ft.) to 300 meters (1000 ft.). Antenna base resistance and reactance values, with estimated tuner parameters, efficiency, and range.

In Table 12.1-1 about vertical antennas, I have given the specific type of antenna tuner necessary and indicated the configuration by a number. Each number indicates a type of “L” (or “Ell”) antenna tuner. I have used these specific tuners because they are the most simple, have the lowest loss, are easiest to tune, have the best inherent bandwidth, and will tune any type of unbalanced vertical antenna. They come in four possible configurations indicated in Figure 11.81-1. The range of values of inductance and capacitance given in Table 12.1-1 may be used with most similar antennas operating near the same frequencies.

12.2 A 140 foot top-loaded vertical

12.21 Configuration. Comments here will reference the drawing TOPHAT3 found at the end of the book. It is a 140 foot tall (42.7 meter) top loaded vertical antenna having a diameter of about 1 foot (0.3 meters). Guys are shown in a separate drawing called GUYS3. The antenna also has four orthogonal 66 foot long (20 meter), highly insulated, 3 x 10 gauge copper coated steel wires attached to the tower top. These wires droop at an angle of 45 degrees. Please note that they are not used as guy wires. An inductor, may, or may not, be attached between the top loading wires and the main tower structure in order to improve the antenna efficiency. Using such a top loading inductor may or may not improve the range of the vertical antenna depending on improvement in efficiency and improved RF current in the vertical portion of the antenna.

12.22 Top Hat. A top loaded vertical antenna can be quite effective at low and medium frequencies. Numerous other configurations of capacitive top loading may be attached to a short vertical antenna to make it appear more like a quarter wave vertical. Top hat wire droop must be limited to roughly 25% to 50% of tower height to avoid significant reversed current vector signal cancellation. A correctly designed top hat will commonly raise the radiation resistance, thus improving antenna efficiency and range.

12.23 Center Loading Coil. Just below the “top hat,” an inductor may be installed to resonate the antenna. This inductor replaces most of the inductance in the antenna tuner at the antenna base. Some serious losses should be expected from this coil. The amount of capacitance of the “top hat” determines the inductance value of the coil. Because RF and lightning voltage at the top of the antenna may be extremely high, this coil needs to be very reliable, have excellent insulation, and have a very high unloaded Q value. Again the Q value must be very high, preferably about 400 or greater.

Configurations of the top hat require that all parts be both strong and corona resistant. Top loading coils tend to make corona more likely. The least bit of corona from the “top hat” elements will ruin the operation of the whole antenna. Moreover corona may not give any indication of its occurrence except for a poor signal in the field.

One corona test is to slowly turn down the transmitter output power level and watch the field strength at a distance of several kilometers. If there is a sudden increase in signal strength or a signal plateau with reduction of power, you may have one or more coronas. Sometimes coronas can be spotted on a dark night with a special camera designed for high tension power lines. However, it is best to use corona rings and rounded ends on all projections similar to a high voltage power line.

Loading coils shown in Table 12.23-2 have a certain amount of reactance left at the bottom of the tower so that an antenna tuner can be used for final adjustment of the antenna to the exact resonant frequency. In areas where wet snow and ice cover such antennas, it may be wise to have an automatic antenna tuner that will correct for small changes in impedance due to weather problems. Of course some further reduction in reliability can be expected from automatic tuners due to their complexity in a very harsh operational environment.

Freq. KHz.	R _{rad.} ohm	X _{base} ohm	R _{loss}	Eff. %	Range for 70 uV. at 1 Kw.
100	.101	-1928	5	1.98%	154 n. mi.
200	.409	-948	5	7.56%	326 n. mi.
300	.934	-614	5	15.7%	430 n. mi.
400	1.70	-441	5	25.3%	494 n. mi.
500	2.73	-332	5	35.3%	534 n. mi.
600	4.07	-255	5	44.8%	558 n. mi.
1500	55.3	+114	5	91.7%	686 n. mi.

Table 12.23 - 1. Calculated impedance, efficiency, and range of a top loaded 140 foot high vertical antenna (V140TH) using a direct connection between the tower top and the top hat.

Freq. KHz	R _{rad.} ohm	X _{base} ohm	X _{coil} ohm	R _{gnd.} ohm	R _{coil} ohm	R _{tot} ohm	Eff. %	Range for 70 uV. @ 1 KW
100	.220	-247	3100	5	7.75	12.97	1.7%	136 n. mi.
200	.779	-267	1400	5	3.5	9.28	8.4%	343 n. mi.
300	1.73	-176	900	5	2.25	8.98	19.3%	467 n. mi.
400	2.85	-165	600	5	1.5	9.35	30.5%	528 n. mi.
500	4.10	-258	400	5	1.0	10.1	40.6%	559 n. mi.
600	4.16	-242	0!	5	0.0	9.16	45.4%	560 n. mi.

Table 12.23 - 2. Calculated impedance, efficiency, and range of a 140 foot top loaded vertical antenna (V140TH) running 1 KW with loading coil of Q=400 between the tower top and the top hat.

Freq. KHz	R _{rad.} ohm	X _{base} ohm	Ind. uHy	Diam in.	Len. in.	Wire Dia. in.	Space wire to wire. in.	Coil Turns
100	.218	-180	3100	36	60	.5	.695	85
200	.836	-136	1400	36	36	.5	.742	48
300	1.60	-223	900	36	36	.5	.921	38
400	2.55	-221	600	36	36	.5	1.122	31
500	3.78	-192	400	36	36	.5	1.364	25
600	5.16	-300	0	0	0	0	0	0

Table 12.23 - 3. Estimated loading inductor parameters for 140 ft. top loaded vertical. Inductor located between tower top and top hat. A small tuning reactance remains at the antenna base to allow final resonating from the ground.

Remarkably, I note that adding a loading coil on this antenna does not seem to significantly improve its efficiency between Table 12.23-1 and Table 12.23-2. This appears to be due to losses in the loading coils. The efficiency of this top loaded antenna will be a function of its radiation resistance divided by the sum of all losses and the radiation resistance. If we do not include antenna tuner losses and calculate the antenna base power only, table 12.23-2 will give a reasonable estimation of the efficiency and 70 microvolt range of such a 140 foot top hat antenna running at the 1000 Watt level.



Ed Note: This is just a “sample chapter” from Robert’s excellent book. If you enjoyed this chapter and would like to obtain the entire book, it is available here in the antennex Shopping Shack BookShelf. [Just click here to go there.](#)

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